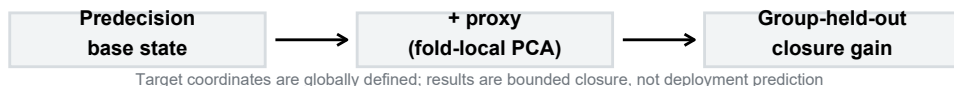


Fold-local proxy reduction preserves the hidden-state closure advantage

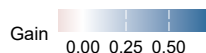
a



b

Proxy-specific closure gains across checkpoints

Each cell is augmented minus base performance

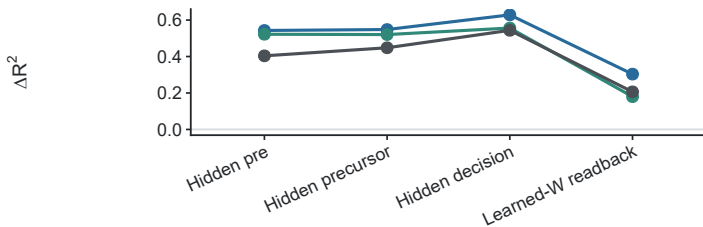


	Qwen				Llama				Gemma			
Random K	-0.02	-0.02	-0.02	-0.03	-0.01	-0.00	0.02	-0.00	-0.01	-0.00	-0.02	-0.00
Gaussian readback	0.30	0.23	0.34	0.30	0.16	0.04	0.29	0.05	0.17	0.02	0.15	0.01
Row-permuted-W	0.30	0.30	0.41	0.39	0.18	0.03	0.27	0.04	0.21	0.02	0.22	0.02
Learned-W readback	0.30	0.30	0.41	0.39	0.18	0.03	0.27	0.04	0.21	0.02	0.22	0.02
H-shape	0.25	0.65	0.28	0.73	0.07	0.24	0.15	0.30	0.09	0.10	0.06	0.08
Direct hidden flow	0.45	0.37	0.44	0.47	0.34	0.07	0.51	0.10	0.28	0.04	0.23	0.03
Hidden decision	0.63	0.42	0.45	0.55	0.56	-0.14	0.53	-0.04	0.54	0.04	0.24	0.02
Hidden precursor	0.55	0.41	0.45	0.51	0.52	0.00	0.53	0.05	0.45	0.04	0.24	0.03
Hidden pre	0.54	0.41	0.45	0.52	0.52	0.06	0.53	0.10	0.40	0.04	0.24	0.03
	Future-state R ²	H-shape R ²	Mechanism F1	Δ U R ²	Future-state R ²	H-shape R ²	Mechanism F1	Δ U R ²	Future-state R ²	H-shape R ²	Mechanism F1	Δ U R ²

c

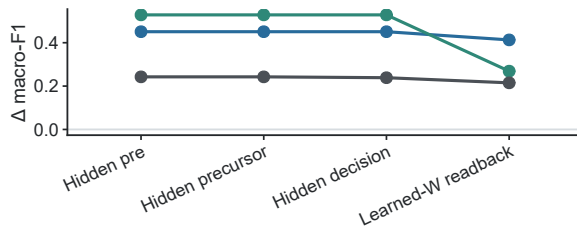
Future-state closure gain

Qwen Llama Gemma



d Mechanism-label gain

Qwen Llama Gemma

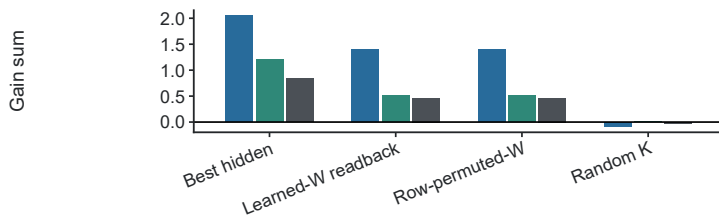


e

Secondary composite summary

Sum of four metric-specific gains

Qwen Llama Gemma



f

Bounded result

Hidden-state proxies add more closure than readbacks

Global target PCA; other audits remain open